Supplementary Material for: `MedCaseReasoning`: Evaluating and learning diagnostic reasoning from clinical case reports

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This document contains supplementary tables, figures, and prompts referenced in the main paper, "`MedCaseReasoning`: Evaluating and learning diagnostic reasoning from clinical case reports," submitted to the Pacific Symposium on Biocomputing (PSB). It is hosted separately from the main manuscript per PSB's policy that supplemental material may be referenced by URL rather than included in the submitted PDF.

Keywords: Supplementary material.

1. Supplementary Tables, Figures, and Prompts

Question	Physician Agreement
Q1: There are no hallucinated details in the case prompt or diagnostic reasoning.	$98.0\%_{95.0\text{–}100.0\%}$
Q2: The final diagnosis is faithful to the article and reasonably inferrable from the details in the case prompt.	$92.0\%_{86.0\text{–}97.0\%}$
Q3: The diagnostic reasoning steps are faithful to the case report and are clinically relevant.	$93.0\%_{88.0\text{–}98.0\%}$

Table 1. Physician agreement rates with 95% confidence intervals for diagnostic prompt evaluation, with $N = 100$ total case reports reviewed. We find that diagnostic cases generated from case reports contain low rates of hallucinations and are faithful to the original reports.

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Journal Name	Prevalence (%)
Journal of Medical Case Reports	8.20%
Clinical Case Reports	7.57%
International Journal of Surgery Case Reports	6.81%
Radiology Case Reports	6.36%
Journal of Surgical Case Reports	2.88%
JAAD Case Reports	2.47%
SAGE Open Medical Case Reports	2.16%
Journal of Orthopaedic Case Reports	2.10%
Case Reports in Medicine	1.87%
European Heart Journal: Case Reports	1.60%

Table 2. Top 10 medical case report journals by prevalence in the PMC Open Access Subset dataset. In total, there are 813 unique journals in `MedCaseReasoning`, spanning 30+ different medical specialties across multiple countries.

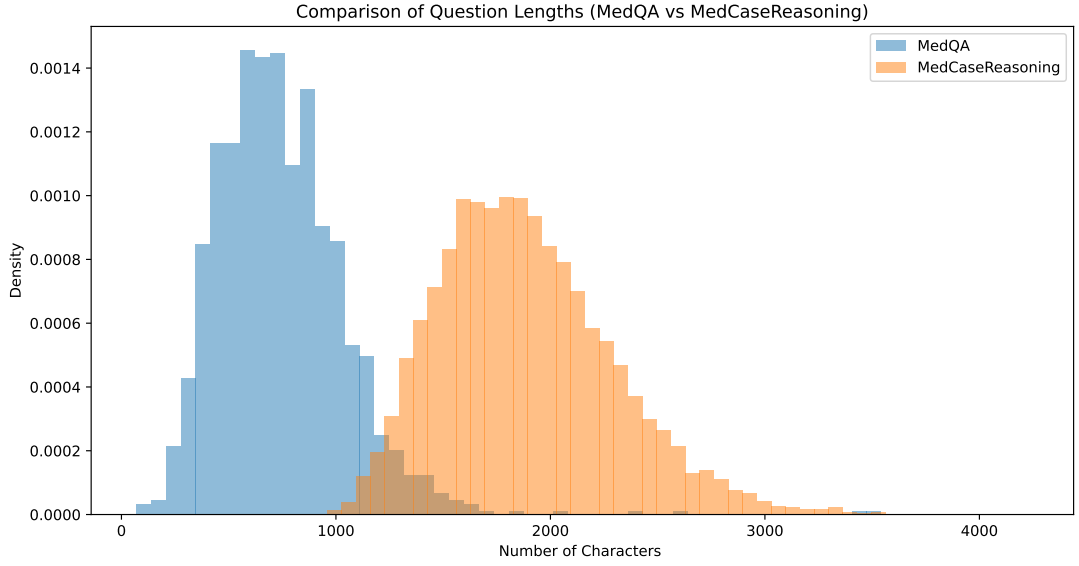


Fig. 1. Comparison of length of questions from MedQA vs. `MedCaseReasoning`. Diagnostic case prompts are, on average, 2.5x longer in `MedCaseReasoning` and contain real patient information vs. synthetic case vignettes.

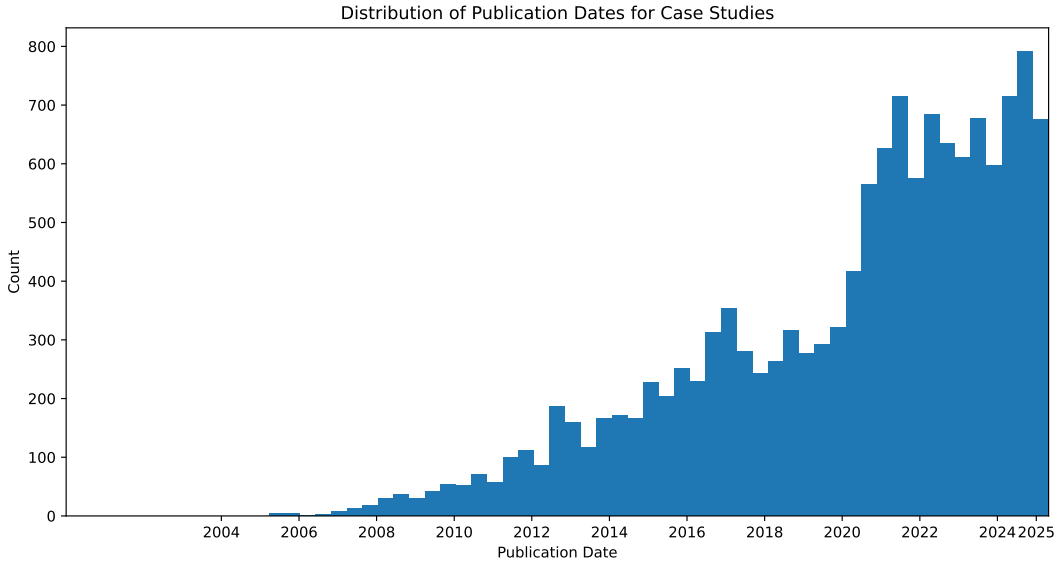


Fig. 2. Distribution of dates of publication for PMC case reports used in `MedCaseReasoning`. Cases are largely from recent dates (after 2020), and over 500 cases are after Jan 1 2025.

NEJM CPC Generalization			
Model	*P@1*	*P@5*	*P@10*
OpenAI o3	**0.430** [0.377–0.490]	**0.579** [0.526–0.639]	**0.623** [0.569–0.675]
DeepSeek R1	0.272 [0.225–0.318]	0.407 [0.354–0.467]	0.437 [0.384–0.497]
QwQ-32B	0.182 [0.136–0.225]	0.281 [0.232–0.334]	0.341 [0.288–0.397]
MedReason-8B	0.169 [0.126–0.215]	0.262 [0.212–0.308]	0.295 [0.245–0.348]
LLaMA-3.1-8B	0.070 [0.043–0.099]	0.142 [0.103–0.182]	0.212 [0.165–0.255]
m1-7B-23k	0.099 [0.066–0.136]	0.172 [0.136–0.215]	0.209 [0.162–0.252]
Qwen-2.5-7B	0.070 [0.043–0.099]	0.109 [0.076–0.142]	0.123 [0.086–0.159]
Fine-tuned on `MedCaseReasoning`			
MedReason-8B (SFT)	**0.179** [0.136–0.225]	**0.278** [0.228–0.328]	**0.348** [0.295–0.404]
LLaMA-3.1-8B (SFT)	0.169 [0.126–0.212]	0.255 [0.205–0.305]	0.308 [0.255–0.358]
Qwen-2.5-7B (SFT)	0.129 [0.093–0.166]	0.202 [0.159–0.248]	0.252 [0.205–0.305]

Table 3. Generalization on NEJM CPC cases (N=302) from.[1] Fine-tuning on `MedCaseReasoning` improves performance on NEJM CPC.

Diagnostic Reasoning

(1) Benign soft-tissue tumors (lipoma or fibroma) were considered but deemed unlikely because "Lipomas and fibromas typically present as solid masses and are rarely associated with infection."

(2) A canal of Nuck cyst was considered but excluded because "The canal of Nuck represents a cystic lesion extending into the inguinal region, which was ruled out based on imaging."

(3) Bartholin's gland abscess was supported by pelvic imaging showing a contrast-enhancing fluid collection at the gland sites; therefore, a diagnosis of bilateral Bartholin's gland abscesses was made.

Stitched Reasoning

Lipomas and fibromas typically present as solid masses and are rarely associated with infection, making benign soft-tissue tumors such as lipomas or fibromas unlikely.

The canal of Nuck represents a cystic lesion extending into the inguinal region, which was ruled out based on imaging.

Pelvic imaging showed a contrast-enhancing fluid collection at the gland sites, supporting the diagnosis; therefore, a diagnosis of bilateral Bartholin's gland abscesses was made.

Table 4. Example of converting enumerated diagnostic reasoning into a cohesive reasoning trace. The fine-tuned model (*LLaMA-3.1-8B-Instruct*) stitches reasoning points without introducing additional information.

Article Title

Sebaceous carcinoma of the breast predominantly characterized by intraductal growth: a case report

Journal

Surgical Case Reports

Article Link

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7040145/

Case Prompt

A 47-year-old Japanese woman with no breast symptoms was referred after screening mammography revealed clustered pleomorphic calcifications in the left breast. Her mother had bilateral breast and ovarian cancer. On examination, there was no palpable mass or nipple discharge. Serum tumor markers (CEA, CA15-3, NCC-ST-439, BCA225) were within normal limits. Breast ultrasound showed a 13×12×7 mm irregular, hypoechoic mass with clear margins and internal high-echogenic foci in the left breast. MRI demonstrated a 14×11×12 mm lesion with early arterial enhancement in the same region. PET-CT revealed focal uptake (SUVmax 3.54) in the left breast without abnormal uptake elsewhere. Core-needle biopsy of the mass showed ductal carcinoma in situ, nuclear grade 2, negative for ER, PgR, and HER2, with a Ki-67 labeling index of 32.4%. Given her family history, hereditary breast and ovarian cancer syndrome was considered, but the patient declined genetic testing. The patient elected to undergo total left mastectomy with sentinel lymph node biopsy.

Diagnostic Reasoning

1. Consideration of invasive ductal carcinoma — "SC is characterized by lobular forms or nests of tumor cells that exhibit sebaceous differentiation, which distinguishes SC from invasive ductal carcinoma (IDC)."

2. Consideration of glycogen-rich clear cell carcinoma — "Glycogen-rich clear cell carcinomas typically become periodic acid-Schiff (PAS)-positive and therefore can be distinguished from SC cells because they do not produce glycogen."

Final Diagnosis

Sebaceous carcinoma

Table 5. A single example from `MedCaseReasoning` of a case prompt, extracted diagnostic reasoning, and final diagnosis.

Prompt 1: Converting Case Reports to Diagnostic QA with Reasoning

You are an expert clinician–educator.
Your job is to **transform a published case report into a teaching diagnostic case** that medical students can work through step-by-step.
In terms of style, think of NEJM's Clinicopathologic Conferences ("Case Records of the Mass General Hospital") as a template.

RULES (Read Carefully—No Exceptions)

---

(1) **Source Fidelity** – Extract facts *only* from the supplied case report.
- Do **NOT** invent, embellish, or "smooth out" missing data.
- Paraphrase narrative prose into concise bullets where helpful, but never add new facts.

(2) **Structure the Teaching Case in Three Phases**
Case Presentation → *Diagnostic Reasoning* → *Final Diagnosis*

(3) **Use the XML Tags Exactly as Shown**
- `<think>...</think>` – your hidden analytic notes (not visible to students).
- `<case_prompt>...</case_prompt>` – the information given to students *before* they generate a differential.
- `<diagnostic_reasoning>...</diagnostic_reasoning>` – numbered bullet reasons, each built as a full sentence followed by a **direct quote**.
- `<final_diagnosis>...</final_diagnosis>` – **single disease/entity name only**, nothing more.

(4) **What Goes Inside `<think>`**
1. *Key points* – What makes this case non-trivial or pedagogically interesting? This should guide where the breakpoint should be.
2. *Ideal breakpoint* – What details of the case presentation should you include and exclude so that students have enough data to reason, but no spoilers?
3. *Author's analytic distinctions* – How did they separate the final diagnosis from look-alikes and other conditions?

(5) **What Goes Inside `<case_prompt>`**
- Present *only* the facts known **before** a working differential was made: chief complaint, HPI, vitals, physical exam, and early investigations.
- Do not include references to Figure 1, Table 1, etc. directly. Summarize any imaging findings from what is given in the text.
- Present the case in the order presented in the case report (e.g., physical labs before imaging, etc.)
- Omit any wording that directly states or hints at the final diagnosis.
- Present this as closely as possible to the style in which the case report is written.

(6) **What Goes Inside `<diagnostic_reasoning>`**
- Numbered list (1., 2., 3., . . .).
- Include every alternative diagnosis mentioned in the article, along with reasons why it was considered and excluded, if provided.
- Each entry: *concise summary of reason* ["direct quote from article"].
- You can use ellipses (. . .) to shorten the quote if there are irrelevant details.
- Quotes must justify why one possibility rose or fell in likelihood.

(7) **What Goes Inside `<final_diagnosis>`**
- **Single disease/entity name** (e.g., `sarcoidosis`).
- No adjectives, punctuation, or explanatory text.

OUTPUT TEMPLATE (copy exactly)

```
<think>
1. [Core tension]
2. [Best breakpoint of case report, what to include and what to exclude]
3. [Key analytic distinctions between competing diagnoses (taken from case report)]
</think>
<case_prompt>
[Your case presentation text, faithful to the report and stopping at the breakpoint]
</case_prompt>
<diagnostic_reasoning>
1. Summarized reasoning - "Direct quote from article..."
2. Summarized reasoning - "Direct quote from article..."
3. ...
</diagnostic_reasoning>
<final_diagnosis>
DiseaseName
</final_diagnosis>
```

SUPPLIED CASE REPORT

<case_report>

$case_report$

</case_report>

Prompt 2: Grading the Quality of Case Reports

You are an expert medical educator tasked with evaluating case reports for their diagnostic-reasoning value. The goal is to find case reports that have a similar style to diagnostic teaching cases from medical textbooks.

CASE REPORT EVALUATION RUBRIC

» HOW TO USE

(1) Read the entire case once without scoring.
(2) Re-read, taking notes.
(3) Inside `<think>...</think>`, write the reasoning that leads you to each score.
(4) Output only the five XML tags shown after the rubric—nothing else.

1. THOROUGHNESS OF CASE PRESENTATION (1–5 points)
Look for: HPI, past history, meds, allergies, vitals, focused exam, labs, imaging, hospital course, outcome.

- **1** – Seriously deficient (identifiers only; no vitals).
- **2** – Major gaps (HPI + vitals OR exam, not both).
- **3** – Adequate (present but sketchy details).

- **4** – Very good (complete data, clear timeline).
- **5** – Exemplary (serial data and course, high quality).

2. EXPLICIT DIFFERENTIAL DIAGNOSIS (Yes / No)
At least two plausible alternatives? If yes → `Yes`; else → `No`.

3. DEPENDENCE ON INTEGRATIVE CLINICAL REASONING (1–5 points)
Measures need to combine at least two data points (history, labs, imaging, etc.).

- **1** – Trivial: lone clue gives answer.
- **2** – Minimal: one dominant clue.
- **3** – Moderate: must merge two findings.
- **4** – High: three or more clues; requires synthesis.
- **5** – Outstanding: stepwise, complex reasoning.

4. TRANSPARENCY OF DIAGNOSTIC REASONING PROCESS (1–5 points)

- **1** – None (no rationale).
- **2** – Superficial (lists without "why").
- **3** – Adequate (brief pivots).
- **4** – Detailed (stepwise, probabilities).
- **5** – Model (structured, addresses pitfalls).

5. STATED FINAL DIAGNOSIS (Yes / No)
Is diagnosis clearly named? Yes → `Yes`; else → `No`.

Additional Rules:

(1) If the given article is not actually a case report, output `NA` for all scores.
(2) Think through whether the final diagnosis can be reasonably deduced from the case presentation when determining the educational value in #3.

OUTPUT TEMPLATE (leave tags exactly as written)

```
<think>
...your internal reasoning for each item...
</think>
<case_presentation_score>[1-5]</case_presentation_score>
<differential_diagnosis_score>[Yes/No]</differential_diagnosis_score>
<integrative_reasoning_score>[1-5]</integrative_reasoning_score>
<transparency_score>[1-5]</transparency_score>
<final_diagnosis_score>[Yes/No]</final_diagnosis_score>
```

SUPPLIED CASE REPORT

```
<case_report>

{case_report}

</case_report>
```

Prompt 3: Detecting Hallucinations and Inconsistencies in Generated Case Prompts

You are the Editor.
Your job is to audit a draft teaching case that was produced from a published case report.
You must confirm strict compliance with all instructions, detect hallucinations, and ensure pedagogic quality.

YOUR INPUTS

---

(1) The original draft you must audit appears between
`<generated_case>...</generated_case>`.
(2) The source article appears between
`<case_report>...</case_report>`.

Here is the original guidelines the draft was produced in accordance with:
`{convert_case_report_prompt}`

CHECKLIST — FAIL ANY ITEM → RAISE A FLAG

A. Source Fidelity ☐ Every fact in each section is traceable to the source article.
☐ No invented details or embellishments.
B. Case Presentation Quality ☐ All facts from the case prompt are present in the source article.
☐ Contains only information known *before* the clinicians formed a differential.
☐ Does **not** reveal the final diagnosis (there should be room for at least some inference).
☐ Provides *sufficient* data (HPI, vitals, exam ± initial tests) for clinicians to formulate a reasonable differential and get the correct final diagnosis.
C. Diagnostic Reasoning Section ☐ Each numbered entry starts with a summary of the reasoning **plus** a direct quote from the article.
☐ Quotes are verbatim or use ellipses (. . .) without changing meaning.
☐ Paraphrased quotes are okay, as long as they retain the original meaning.
☐ **Rationales reference only information that already appears in `<case_prompt>`** (not based on new findings, confirmatory tests, or data withheld from students).

D. Final Diagnosis Tag ☐ Final diagnosis is *reasonably* deducible from the case-presentation facts.
— i.e., the final diagnosis should not depend entirely on some test, imaging, or lab result not given in the case presentation.
E. No Hallucinations Anywhere ☐ Every datum, quote, or diagnosis is found in the case report.

HOW TO REPORT YOUR FINDINGS

Output **only** the two XML blocks below.

(1) `<flags>...</flags>`
- If an item fails, add a line `FLAG: [short descriptor]`.
- Use one line per failed item, drawn from this controlled vocabulary:
`CASE_PROMPT_HALLUCINATION`, `FINAL_DIAGNOSIS_IN_CASE_PROMPT`,
`INSUFFICIENT_INFO_FOR_DIAGNOSIS`, `DIAGNOSTIC_REASONING_HALLUCINATION`, `OTHER`.
- If **no** issues, write `NONE`.

(2) `<editor_comments>...</editor_comments>`
- Briefly justify each flag (one sentence each).
- If no flags, you may omit or leave empty.

Example when problems exist:

```
<flags>
FLAG: SOURCE_FIDELITY
FLAG: REASONING_EXTRA_INFO
</flags>
<editor_comments>
- SOURCE_FIDELITY: Mentions "family history of SLE," not present in article.
- REASONING_EXTRA_INFO: Rationale cites a biopsy result that is not included in the case_prompt.
</editor_comments>
```

Example when everything passes:

```
<flags>
NONE
</flags>
<editor_comments></editor_comments>
```

INPUT BLOCKS TO REVIEW

Here is the reference case report:

```
<case_report>
{case_report}
</case_report>
```

Here is the diagnostic case generated by the model:

```
<case_prompt>

{generated_case_prompt}

</case_prompt>

<diagnostic_reasoning>

{generated_diagnostic_reasoning}

</diagnostic_reasoning>

<final_diagnosis>

{generated_final_diagnosis}

</final_diagnosis>
```

Prompt 4: Stitching Reasoning Trace From Enumerated Diagnostic Reasoning

```
Your job is to convert a list of diagnostic reasoning points into a cohesive diagnostic reasoning narrative.


INPUTS
* CASE_PROMPT:
{case_prompt}

* REASONING_POINTS (in the order they were generated):
{reasoning_points}

* FINAL_DIAGNOSIS:
{final_diagnosis}


TASK
Write a single, cohesive reasoning trace by stitching together all the REASONING_POINTS.
The REASONING_POINTS are given as an enumerated list, where each point is a brief summary of a particular reasoning point,
followed by a quote from the full case report that supports the reasoning point.
This should be written from the perspective of a reasoning trace that an LLM chatbot would write in response to a case
presentation.
```

```
Do not use the past-tense nature of REASONING_POINTS; instead make them present-tense and third-person as you are considering
each potential diagnosis.


OUTPUT
Only use the reasoning from the REASONING_POINTS to come up with the final diagnosis.
Use ALL of the REASONING_POINTS in producing the reasoning trace.
Do not include your own reasoning over the case; only use the reasoning points provided.
Try to incorporate as much of the quotes as possible into the reasoning trace, using word-for-word copies when possible (do not
put quotation marks in the reasoning trace).
You may rephrase the reasoning points, but only for style and tone, not for substance.
No headings, no bullets, no numbered lists--just a continuous explanatory narrative.

Place this between the tags <stitched_reasoning> and </stitched_reasoning>.
```

Prompt 5: Grading Reasoning Recall

```
You are an experienced medical expert tasked with comparing diagnostic reasoning statements that support a given diagnosis for
a given patient case.
Your goal is to find supporting statements in the predicted diagnostic reasons that match the groundtruth diagnostic reasons.

For each of the statements in Groundtruth Diagnostic Reasons, you need to find the statement or statements in the Predicted
Diagnostic Reasons that state the equivalent justification for the diagnosis.
For instance, if the groundtruth diagnostic reason is "The patient has a fever", and the predicted diagnostic reason is "The
patient has a fever due to a viral infection", then this is a match.
If the groundtruth diagnostic reason is "The patient has a fever", and the predicted diagnostic reason is "The patient has a
sore throat", then this is not a match.


Instructions:
1. Analyze each statement in Groundtruth Diagnostic Reasons.
2. For each statement in Groundtruth Diagnostic Reasons, find any matching statements in Predicted Diagnostic Reasons.
3. Create a JSON object with the following structure:
- The main key should be "matching_dict"
- Each key within "matching_dict" should be a number representing a statement from Groundtruth Diagnostic Reasons
- The value for each key should be a list of matching statements from Predicted Diagnostic Reasons
- If there are no matches for a statement, use an empty array


Before providing your final output, wrap your analysis inside <diagnostic_comparison> tags:
1. List all statements from Groundtruth Diagnostic Reasons and Predicted Diagnostic Reasons.
2. For each statement in Groundtruth Diagnostic Reasons, consider potential matches from Predicted Diagnostic Reasons:
- List pros and cons for each potential match
- It's OK for this section to be quite long
3. Summarize your final matching decisions
4. In the JSON output, only include the statements that are in the Predicted Diagnostic Reasons.
5. In the JSON output, the statements should appear exactly as they are in the Predicted Diagnostic Reasons, verbatim, letter
for letter. Do not modify the statements in any way.

Wrap your JSON output in "'json tags.

Example of the required JSON structure:
"'json
{
"matching_dict": {
"1": [],
"2": ["Matching statement 1", "Matching statement 2"],
"3": ["Matching statement 3"]
}
}
"'
```

Prompt 6: Diagnostic Question Template

```
Read the following case presentation and give the most likely diagnosis.
First, provide your internal reasoning for the diagnosis within the tags <think> ... </think>.
Then, output the final diagnosis (just the name of the disease/entity) within the tags <answer> ... </answer>.

----------------------------
CASE PRESENTATION
----------------------------
{case_presentation}

----------------------------
OUTPUT TEMPLATE
----------------------------
<think>
...your internal reasoning for the diagnosis...
</think>
<answer>
```

```
...the name of the disease/entity...
</answer>


a_prompt = """<think>
{reasoning}
</think>

<answer>
{answer}
</answer>
"""
```

Prompt 7: Diagnostic Accuracy LLM-as-a-judge

```
Is our predicted diagnosis correct (y/n)?
Predicted diagnosis: {predicted_diagnosis}, True diagnosis: {actual_diagnosis}
Answer [y/n].
```

References

1. D. McDuff, M. Schaekermann, T. Tu, A. Palepu, A. Wang, J. Garrison, K. Singhal, Y. Sharma, S. Azizi, K. Kulkarni *et al.*, Towards accurate differential diagnosis with large language models, *Nature* , 1 (2025).